

CS111 Project

Project Description

Students are required to form a team of four members and choose one of the following system ideas to design and implement a complete system, as specified in the requirements section.

1. Smart Parking System
2. Event Ticket Booking System
3. Online Food Ordering System
4. Fitness Class Booking System
5. Warehouse Inventory System
6. Digital Wallet Management System
7. Taxi/Car Ride Booking System

Project Objectives

After implementing the term project, students will practice:

- Design and implement class inheritance hierarchies in JAVA.
- Explain the fundamental concepts of OOP: encapsulation, inheritance, abstraction and polymorphism.
- Use arrays of objects and arraylists.
- Understand and utilize the exception handling mechanisms.
- Create and use text or binary files in java
- Practice teamwork, documentation, and presentation.

Project Requirements

- Each class within the project should contain, as a minimum, an empty and a parameterized constructor, the setters/getters methods, and a method to print all the details.
- All classes should apply the encapsulation concept.
- The fields data must be inserted by the user and verified, when needed, in the corresponding methods, such as numeric values must be valid numbers and logical constraints must be enforced (e.g., availability, balance, capacity).
- You must create a text file where all the data of the objects created in the main will be saved.
- In this project, you must use the concepts of CS 111 in your solution: (Inheritance, Polymorphism, Abstract classes, Interface, Composition, ArrayList, exception handling (including built-in exceptions and at least one custom exception), and Files)

- The main class must display a menu of options in the console. The program must run continuously using a loop until the user selects the Exit option. All invalid menu selections must be handled properly.
- All classes and important methods must be documented using Java documentation comments (Javadoc).
- The system must handle invalid inputs gracefully and must not crash when users enter incorrect or unexpected values.

Project Submission

Phase 1 – Due Date: Thursday, 16 April 2026, 11:59 PM

- **Report file** (template provided), including:
 - Project overview
 - UML class diagram showing all classes and their relationships

Phase 2 – Due Date: Thursday, 7 May 2026, 11:59 PM

1. **Compressed folder** containing the complete project source code
2. **Report file** (template provided), including:
 - Project overview
 - UML class diagram
 - Snapshot of the program execution trace

***Note:** Minor modifications to the UML class diagram are allowed in Phase 2; however, a clear justification must be provided.*

Project Presentation – Week 14

- Each team must prepare a presentation.
- All team members must participate in the presentation.
- Teams must be prepared to answer questions related to their implementation.
- **Presentation Contents:**
 - The project title, your names, IDs and your section
 - Project Overview
 - Purpose of the system
 - Problem the system solves
 - Target users
 - System Features
 - Main functionalities
 - Menu options
 - Class Diagram
 - OOP Concepts Used
 - Mention the main OOP concepts used and include brief code snapshots demonstrating their application. (*Do not include excessive details.*)

- Live Demonstration
 - Run the program
 - Show key features
- Conclusion
 - Project summary
 - Objectives achieved
 - Challenges & Lessons Learned
- **Presentation format requirements:**
 - The presentation should be simple, precise, well-organized, and visually clear.
 - Slides must be prepared using PowerPoint or Google Slides
 - Font type: Arial, Calibri, or Times New Roman
 - Font size:
 - Title: 32–40 pt
 - Section headings: 24–28 pt
 - Body text: 18–22 pt
 - Text color: Dark (black or dark blue) on a light background
 - Maximum 6–7 bullet points per slide
 - Use bullet points, not long paragraphs
 - Code screenshots should be clear and readable
 - Consistent slide design and layout throughout the presentation
 - Presentation duration: 15 minutes per team

Important Notes

- All programs must be compiled and run correctly.
- Copying or plagiarism is strictly prohibited.
Any team found copying another solution will receive ZERO marks.